ERDŐS–STRAUS

A proof dependency map — certified restrictions and the remaining existence gap

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Lean 4.19.0 / Mathlib v4.19.0

ES CONJECTURE — OPEN

For every integer $n \geq 2$, positive integers x, y, z satisfy $4/n = 1/x + 1/y + 1/z$.

Both conditional implications are kernel checked

H_D — DYADIC EXISTENCE

Every prime p ≡ 1 (mod 4) has a dyadic Type II source.
Unproved; this bridge route is now inactive.

H_N — NONRESIDUE EXISTENCE

Every prime p ≡ 1 (mod 4) has a nonresidue denominator inside p/4 < x < 3p/4. Active existence target.

MISSING: a uniform existence / selection theorem

B4 and matching restrict methods; they do not establish source existence.

EXACT ALGEBRA CERTIFIED

TYPE II CONSTRUCTION

Positive source and endpoint triples
B1 integer return ⇔ δt divides k
General character transfer: pending

RESTRICTION CERTIFIED

B4 RETURN + DESCENT

Four rational steps, exact gate
t = 1 and δ odd
No new dyadic endpoints

GLOBAL MATCHING CERTIFIED

INTEGRAL MATCHING

All orientations; triples up to permutation
At most one neighbor; no self-loops
Integral iteration cannot branch

LOCAL THEOREM CERTIFIED

SQUARE-CLASS FORCING

Complement-preserving quotient
Finite energy certificate at 577
Uniform choice of x remains open

COMMON TRUNK

FOUNDATIONS

Integer ES: polynomial ⇔ fraction identity
Scaling and reduction to primes
Open hypotheses are definitions, not axioms

ESTABLISHED FORMAL MATHEMATICS

MATHLIB DEPENDENCIES

Dirichlet and quadratic reciprocity
Groups, divisibility and finite sets
Imported proofs; exact versions pinned

NOT CERTIFIED BY THIS RUN

REMAINING TRANSLATION

General B1 character application
Complete finite obstruction files
Uniform analytic existence bound

HOW TO READ THIS MAP

- Kernel checked
- Imported formal theorem
- Partial / not yet formalized
- Open universal assertion

Solid arrows: established dependencies or conditional implications. Dashed arrows: a missing existence / selection step.

A certified conditional theorem does not prove that its premises occur. The companion ledger lists exact Lean statements and remaining gaps.